

Structured Query Language (SQL)

Introduction

SQL (Structured Query Language) is a programming language used to store, retrieve, manipulate, and manage data in relational databases.

Popular databases that use SQL include MySQL, PostgreSQL, Microsoft SQL Server, Oracle

General Database Structure

1.Database Software:

It can be regarded as common database software, such as MySQL,PostgreSQL, MsSQL, etc.

2.Database:

A **database** is an organized collection of data that can be easily stored, accessed, and managed.

3. Schema:

This is a collection of Tables, which can be split and grouped according to logic, and some table details meta settings can be set on this layer, like a blueprint.

4. Table:

The key section of the database. Tables record different data entities, such as users, products, etc.

5. Data:

The most valuable part of the database is the data itself. Different types of data can be stored.

Different types of MySQL

There are two main types in MySQL

- **DDL**(Data definition language)
- **DML**(Data Manipulation language)
- **TCL** (Transaction Control Language)

Data definition language

DDL (Data Definition Language) is used to create, modify, and delete the structure of database objects, such as tables, databases, and indexes.

	Core commands	purpose
1	Create	makes new tables
2	Alter	modifies table structures
3	Drop	deletes an entire table permanently)
4	Truncate	removes all records but keeps the structure
5	Rename	changes the name of a database object

Data Manipulation language

DML (Data Manipulation Language) is used to insert, modify, and delete data inside database tables.

	Core commands	purpose
1	Insert	adds new rows
2	Update	edits existing data
3	Delete	removes specific records
4	Select	retrieves data

Transaction Control Language

TCL (Transaction Control Language) is used to **manage transactions in a database**. A transaction is a group of SQL operations treated as a single unit

	Core commands	purpose
1	Commit	permanently saves the changes

2	Rollback	undoes changes made in the current transaction
3	Savepoint	creates a point within a transaction

Datatypes of SQL

SQL DATA TYPES	01 NUMERIC	INT, DECIMAL, FLOAT, NUMERIC
	02 BINARY	BINARY, BLOB
	03 STRING	CHAR, VARCHAR, TEXT
	04 BOOLEAN	BOOLEAN
	05 DATE/TIME	DATE, TIME, DATETIME

Movie Database Management and Analytical Reporting Case Study:

#Create a database and use the database

```
create database Movies_DB;

use Movies_DB;

create table movie
(
  rel_year int,
  mov_name varchar(100),
  act_name varchar(50),
  lang varchar(10),
  dir_name varchar(100)
);

INSERT INTO MOVIE VALUES
(1960,'Kalaithur Kannamma','Kamal Haasan','Tamil','A. Bhimsingh'),
(1962,'Kannum Waralum','Kamal Haasan','Malayalam','K. S. Sethumadhavan'),
(1962,'Paadha Maanikkai','Kamal Haasan','Tamil','K. Shankar'),
(1962,'Pearithaal Pasi Theerum','Kamal Haasan','Tamil','A. Bhimsingh'),
(1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy'),
(1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan'),
(1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam'),
(1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan'),
(1971,'Anubhavgal Paalichakal','Mammootty','Malayalam','K. S. Sethumadhavan'),
(1972,'Kurathi Megan','Kamal Haasan','Tamil','K. S. Gopalakrishnan'),
(1973,'Arangetrani','Kamal Haasan','Tamil','K. Balachander'),
(1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan'),
(1973,'Sollathaan Ninaikkiren','Kamal Haasan','Tamil','K. Balachander'),
(1974,'Ambu Thangai','Kamal Haasan','Tamil','S. P. Muthuraman'),
(1974,'Aval Oru Thodar Kathai','Kamal Haasan','Tamil','K. Balachander'),
(1974,'Gomastavin Magal','Kamal Haasan','Tamil','A. P. Nagarajan'),
(1974,'Kanyakumari','Kamal Haasan','Malayalam','K. S. Sethumadhavan'),
(1974,'Naan Avanillai','Kamal Haasan','Tamil','K. Balachander'),
(1974,'Panathukkaga','Kamal Haasan','Tamil','M. S. Senthil'),
(1974,'Paruva Koalam','Kamal Haasan','Tamil','Jos A.N. Fernando'),
(1974,'Vishnu Vijayam','Kamal Haasan','Malayalam','N. Sankaran Nair');
```

1, Retrieve all details from the movie table.

```
select * from MOVIES;
```

Solution:

rel_year	mov_name	act_name	lang	dir_name
1960	Kalathur Kannamma	Kamal Haasan	Tamil	A. Bhimsingh
1962	Kannum Karalum	Kamal Haasan	Malayalam	K. S. Sethumadhavan
1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
1962	Paarthaal Pasi Theerum	Kamal Haasan	Tamil	A. Bhimsingh
1963	Anandha Jodhi	Kamal Haasan	Tamil	V. N. Reddy
1963	Vanambadi	Kamal Haasan	Tamil	G. R. Nathan
1970	Maanavan	Kamal Haasan	Tamil	M. A. Thirumugam
1971	Annai Velankanni	Kamal Haasan	Tamil	Thankappan
1971	Anubhavgangal Paalichakal	Mammootty	Malayalam	K. S. Sethumadhavan
1972	Kurathi Magan	Kamal Haasan	Tamil	K. S. Gopalakrishnan
1973	Arangetram	Kamal Haasan	Tamil	K. Balachander
1973	Kaalachakram	Mammootty	Malayalam	K. Narayanan
1973	Sollathaan Ninaikkiren	Kamal Haasan	Tamil	K. Balachander
1974	Anbu Thangai	Kamal Haasan	Tamil	S. P. Muthuraman
1974	Aval Oru Thodar Kathai	Kamal Haasan	Tamil	K. Balachander

2.Display only movie names and actor names.

```
SELECT mov_name, act_name  
FROM movies;
```

Solution:

mov_name	act_name
Kalathur Kannamma	Kamal Haasan
Kannum Karalum	Kamal Haasan
Paadha Kaanikkai	Kamal Haasan
Paarthaal Pasi Theerum	Kamal Haasan
Anandha Jodhi	Kamal Haasan
Vanambadi	Kamal Haasan
Maanavan	Kamal Haasan
Annai Velankanni	Kamal Haasan
Anubhavangal Paalichakal	Mammootty
Kurathi Magan	Kamal Haasan
Arangetram	Kamal Haasan
Kaalachakram	Mammootty
Sollathan Ninaikkiren	Kamal Haasan
Anbu Thangai	Kamal Haasan
Aval Oru Thodar Kathai	Kamal Haasan
Gumasthavin Magal	Kamal Haasan
Kanyakumari	Kamal Haasan

3.Display distinct movie languages.

```
select distinct lang from movies;
```

Solution:

Result Grid		File
	lang	
▶	Tamil	
	Malayalam	
	Telugu	
	Kannada	
	Hindi	
	Bengali	
	Silent	
	English	

4.Retrieve unique actor names.

```
SELECT DISTINCT act_name
FROM movies;
```

Solution:

	act_name
▶	Kamal Haasan
	Mammootty
	Rajinikanth
	Mohanlal
	Vijay
	Silambarasan
	Ajith Kumar
	Suriya
	Trisha
	Nayanthara
	Karthi
	Dhanush
	Dulquer Salm...
	Amala Paul
	Rashmika Ma...
	Manju Warior
	Sai Pallavi
	Parvathy Thi...
	Vijay Sethup...